LECTURE ACTIVITY 2: DON'T GET *RENT* OUT OF SHAPE

STAT 1110: Applied Statistical Concepts and Methods – Fall 2026 with Dr. Ethan P. Marzban

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Learning Objectives:

By the end of this activity, you should be able to

1) Interpret proportions as summary of categorical variables
2) Identify and interpret visuals that are appropriate for categorical data (bar graphs).
3) Describe the distribution of a categorical variable by referring to counts/proportions/percentages.
4) Interpret measures of center (mean, median, mode).
5) Calculate and interpret measures of variability (range, sd and IQR).
6) Identify and interpret visuals that are appropriate for quantitative data (dotplots and histogram).
7) Describe distribution of a quantitative variable by referring to shape, center, spread, and unusual observations.
8) Compare distributions with respect to shape, center and variability between two or more distributions displayed in dotplots or histogram.
9) Make predictions about centers and means, and about variability and standard deviations, based on the context of a variable.
10) Develop critical statistical thinking.

Key Terms:

- Distribution
- Descriptive Statistics
- Bargraph/barplot
- Histogram
- Dotplot
- Scatterplot
- Mean
- Median
- SD (Standard Deviation)
- IQR (InterQuartile Range)

Introduction to the Dataset

In early August, I (Dr. Ethan) went to *Zillow*[1] (a popular online database listing houses and apartments for sale and/or rent) to explore some of the listings for apartments. I only considered properties that were listed as apartments (i.e. I excluded any listings for individual rooms within a shared house) and only considered listings for which information on the following variables was available:

- **NumBed**: The number of bedrooms (Studio apartments are listed as having "0" bedrooms)
- **NumBath**: The number of baths (including half-baths)
- **Size**: The size (in square feet)
- **Price**: The rental price (in USD per month)
- **Pool**: Whether or not the property has a pool (1 = has a pool, 0 = does not have a pool)

The first few rows of the corresponding **Data Table** are as follows:

[1] https://www.zillow.com/

NumBed	NumBath	Size	Price	Pool
1	1.0	185	599	1
1	1.0	93	599	1
1	1.0	122	799	1
1	1.0	122	899	1
1	1.0	185	999	1
2	1.0	151	2800	0
The table continues, and contains 131 rows in total				

Table 1: First Few Rows of the Data Table

Question 1

Describe an observational unit.

Question 2

Based on the data dictionary, classify each of the variables as either quantitative or categorical. For the categorical variables, list the levels.

- **NumBed**: ◯ Categorical ◯ Quantitative
- **NumBath**: ◯ Categorical ◯ Quantitative
- **Size**: ◯ Categorical ◯ Quantitative
- **Price**: ◯ Categorical ◯ Quantitative
- **Pool**: ◯ Categorical ◯ Quantitative

Which Plot When?

We define the **distribution** of a variable to be a description of the values the variable takes and how frequently it takes those values. Distributions are often displayed by way of **descriptive statistics** (graphical and numerical summaries of data).

When producing graphical summaries of variables, we need to pay attention to two main things: the number of variables being described, and the type (categorical/quantitative) of each variable.

- When visualizing the distribution of a single categorical variable, we use a **bargraph** (aka a **barplot**).
  - In a bargraph, we set up as many “bars” (rectangles) as there are levels of the variable. The height of each bar is proportional to the frequency of each level (i.e. how often each level occurs in the observed data).
- When visualizing the distribution of a single quantitative variable, we use a **histogram** or a **dotplot**.
  - In a histogram, we divide the variable up into categories and figure out how many observed values fall within each category. We then set up as many bars as categories, and make the

height of each bar proportional to the frequency of the corresponding category. Dotplots are similar, except instead of bars we simply place a dot in the center of each bin to indicate a count.

- When comparing two quantitative variables, we use a **scatterplot**.
  - In a scatterplot, each point's x-coordinate is dictated by the value of one of the variables being compared and the y-coordinate is dictated by the value of the other variable.

Question 3
For each of the following scenarios, identify the appropriate plot. If there are multiple possible plots, list all of them.

(a) Ronaldo wants to summarize the **NumBath** variable.

(b) Folami wants to summarize the relationship between the number of bedrooms in an apartment and its listed price.

More Plots!

While we're talking about plots, let's get a bit more practice extracting information from plots (like we did in the last Lecture Activity, on Monday). Consider the following plot:

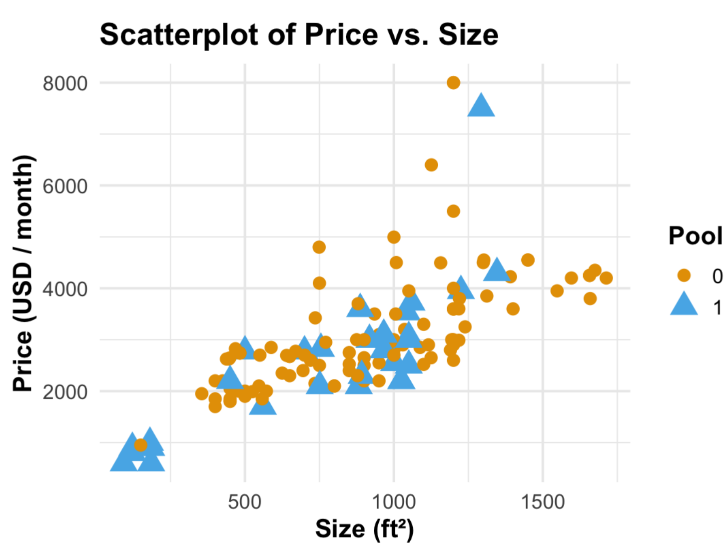


Figure 1: A scatterplot of apartment sizes, prices, and number of bedrooms from a handful of San Luis Obispo rentals.

In scatterplots (like the one in Figure 1 above), each point typically corresponds to an observational unit. In particular:

- The x-coordinate of each point is dictated by the value of a particular variable
- The y-coordinate of each point is dictated by the value of another particular variable.

For example, each point in Figure (1) above represents a particular San Luis Obispo apartment - the x-coordinate is dictated by the size of the apartment, and the y-coordinate is dictated by the price of the apartment. there are several attributes (stylistic choices) we can play around with to encode information from variables. Two commonly-used attributes are:

- **Shape:** we can change the shape of each point depending on the corresponding observational unit's value of a particular variable
- **Color:** we can change the shape of each point depending on the corresponding observational unit's value of a particular variable

Sometimes, we can use one variable to vary across two different attributes (i.e. both color and shape points differently depending on the value of a particular variable).

Question 4
How many variables have information encoded in Figure (1) above? What variables are those?

Question 5
Do larger apartments tend to be more expensive? Does your answer change depending on whether or not the apartment complex has a pool? Justify your answers briefly, referencing specific aspects of the plot as necessary.

Shown below is a plot that contains the same information as displayed in Figure (1), just organized slightly differently (if you're curious this uses a technique called *facetting*, which you can learn more about in more advanced data science courses):

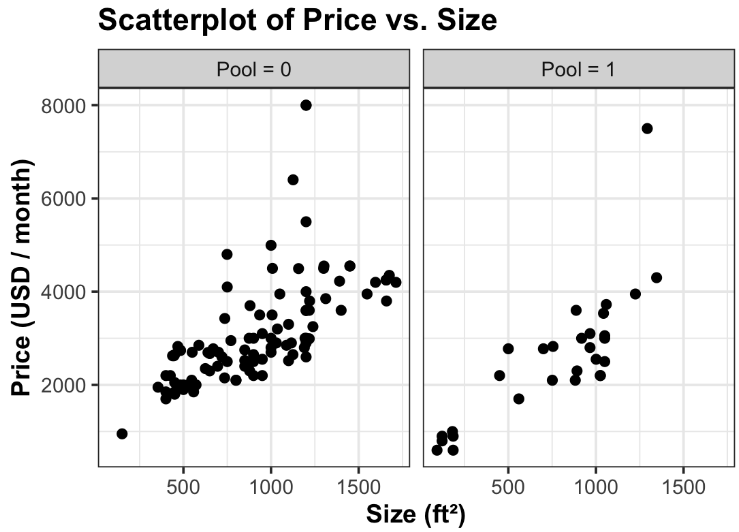


Figure 2: A scatterplot of apartment sizes, prices, and number of bedrooms from a handful of San Luis Obispo rentals.

Question 6

Which plot (Figure 1 or 2) do you think tells the "story" (i.e. your answers to Question 5 above) better? **Note:** there isn't necessarily a right or wrong answer to this question, so please justify your answers!

Numerical Summaries

As mentioned above, descriptive statistics aren't just about graphs - sometimes we need to use *numerical* summaries to describe a distribution. First, let's ease into the math by way of a concept you've likely encountered before: **proportions**. We can define a proportion as:

$$\text{proportion} = \frac{\text{number of "favorable" elements}}{\text{total number of elements}}$$

For example, consider the following list: $\{x, x, y, z, z, z, w, w\}$. The proportion of y's is 1/8 = 0.125 = 12.5% since one of the eight elements was equal to y.

Pro Tip

Proportions can be reported in three different ways: as fractions, as decimals, or as percents.

Numerically Describing Categorical Variables

In the context of a categorical variable, we sometimes use the term **relative frequency** to mean the same thing as a proportion. The **frequency** of a category is the number of observational units in that category, and the relative frequency of a category is simply the category's frequency divided by the total number of observational units.

Question 7

Consider the following plot of the **Pool** variable:

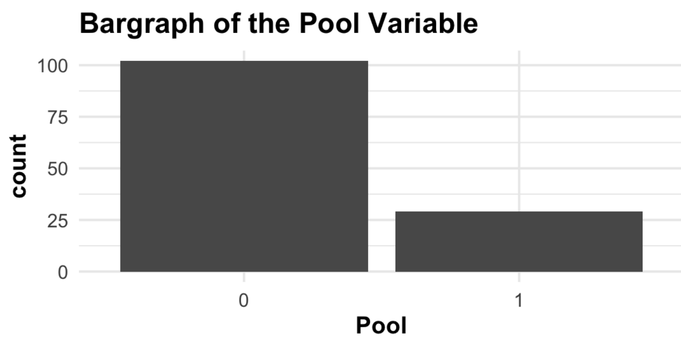


Figure 3: A bargraph of the values in the Pool variable.

Approximately what is the relative frequency of apartments that were located in complexes with pools? Reminder: there were 131 apartments included in the dataset.

Numerically Describing Quantitative Variables

Now let's pivot to talking about how to describe quantitative variables using numerical summaries. Broadly speaking, we have two types of summary measures:

- **Measures of Central Tendency**: these are values that seek to describe "typical," "central," or "common" values of the variable.
- **Measures of Variability**: these are values that seek to describe how "spread out," or how "variable" values of the variable are.

Measures of Central Tendency

There are two main types of measures of central tendency we'll discuss in this class: the **mean** and the **median**. Both of these seek to describe the "central" value of a variable, but they do so in two different ways.

The mean essentially answers the question: where would we need to place a fulcrum underneath a dotplot of the values to make the plot perfectly balanced?

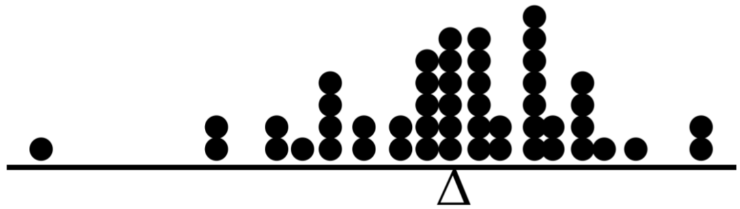


Figure 4: A dotplot with the mean indicated, demonstrating how the mean represents the position of a fulcrum that ensures the entire system is balanced.

The formula to calculate the mean of a list of numbers is:

$$\text{mean} = \frac{\text{sum of all the elements}}{\text{number of elements}}$$

For example, the mean of the list {1, 2, 2, 3, 4, 5} is

$$\text{mean} = \frac{1+2+2+3+4+5}{6} = \frac{17}{6} = \boxed{2.8\overline{3}}$$

The median is essentially the "midpoint" of the data. To calculate the median:

1) Arrange the values in increasing order

2) Cross off the first and last values

3) Cross off the first and last values that haven't been crossed off

4) Continue until either there is only one element left uncrossed (in which case this is the median) or there are two elements left uncrossed (in which case the median is the mean of the two uncrossed values).

For example, the median of the list {1, 2, 2, 3, 4, 5} is

$$\cancel{1} \quad \cancel{2} \quad 2 \quad 3 \quad \cancel{4} \quad \cancel{5} \implies \text{median} = \frac{2+3}{2} = \boxed{2.5}$$

There is another measure of central tendency, called the **mode** which represents the most frequently occurring value in the dataset. It is possible that multiple values occur frequently, in which case we say the dataset is **multimodal**.

Question 8

Calculate the mean and median of the values $\{-1, 0.1, 0.35, 1.44, 2.01, 0.33, 3.22\}$.

- **Mean:** ___________
- **Median:** ___________

Question 9

(a) The mean price of rentals included in the dataset is \$2,994.649 per month. What was the total (sum of all prices) included in the dataset? Recall that there were 131 apartments included in the dataset.

(b) Suppose I incorporated a new (132^{nd}) apartment into the dataset, whose rental price is \$3,000 per month. What is the new mean (of all 132 apartment rental prices)?

Question 10

(a) Will the mean of a list of numbers necessarily be an element of the list?

(b) Under what conditions will the median of a list of numbers be one of the numbers that was observed? **Hint:** think about the number of observations.

Measures of Spread

There are two main types of measures of spread we will use in this class: the **Standard Deviation** (often abbreviated to **SD**) and the **InterQuartile Range** (often abbreviated to **IQR**). The formula for the SD is somewhat complicated and involved, so I won't ask you to use it (yet) - for now, we'll just rely on computers to calculate it for us. The IQR is calculated using the formula

$$\mathrm{IQR} = \mathrm{Q}_3 - \mathrm{Q}_1$$

where Q_1 and Q_3 denote the first and third **quartiles** of the data, respectively. Q_1 is calculated as the median of the values that fall below the median, and Q_3 is calculated as the median of the values that fall above the median. I typically won't ask you to compute Q_1 and Q_3 by hand; instead, these will either be given to you or we'll use a computer to calculate them.

Pro Tip

The mean, median, SD, and IQR will all be in the original units of measurement.

Question 11

The first quartile of rent prices is $2,200 and the third quartile is $3,600. Calculate the IQR of prices included in the dataset. Be sure to include units!

If you're curious, the SD of rents is $1,236.242 per month.

Describing Distributions

Finally, let's put everything together and talk about how to describe distributions of quantitative variables. First: allow me to say a few words on histograms vs. dotplots. Both histograms and dotplots are valid plots to visualize the distribution of a quantitative variable, and both give roughly the same information. Next week, we'll see some situations in which dotplots are slightly preferred over histograms, but for now let's stick with histograms for the rest of today's Lecture Activity. For reference, here is a histogram of the **Price** variable:

Figure 5: A histogram of apartment prices.

When describing a histogram/distribution, we need to describe four things: **Shape**, **Center**, **Variability**, and **Outliers**. Let's discuss each of these separately.

Shape

When describing the shape of a histogram, there are two things to look at:

- **Skew**: is there a "tilt" or a "tail" on the histogram? The direction of the tail dictates what we call the *skew* - if there is no skew, we say the histogram is **symmetric**.
- **Number of Peaks**: how many "peaks" are there on the histogram? We refer to the peaks of a histogram as **modes**.

There are three main possibilities when it comes to skew: right-skew, left-skew, or no-skew (aka symmetric).

Figure 6: Examples of right-skewed, left-skewed, and symmetric distributions.

There are three main possibilities when it comes to the number of peaks: unimodal (one mode), bimodal (two peaks), or multimodal (three or more peaks).

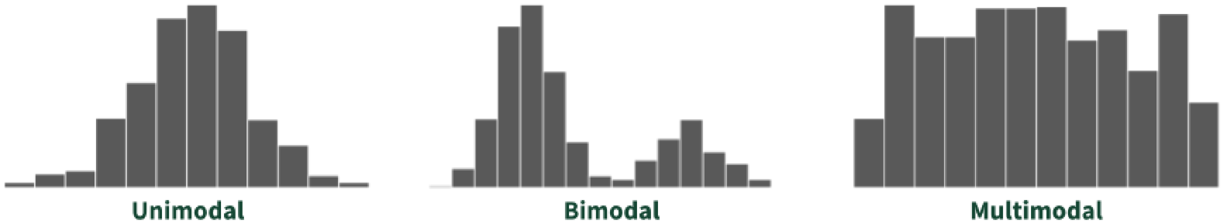


Figure 7: Examples of unimodal, bimodal, and multimodal distributions.

Question 12

Describe the shape of the distribution of prices as displayed in Figure (5) above. Make sure to use the proper terminology introduced above!

Center

As we saw earlier in this activity, there are two main measures of central tendency we use in this class: the mean and the median. For this course, we will use the following convention:

- When the data is skewed, we will use the median to describe the center
- When the data is symmetric, the median and mean will be roughly equal so it doesn't matter which one we use

Question 13

(a) For right-skewed distributions, how will the mean compare to the median?
- ◯ The mean will be smaller than the median
- ◯ The mean will be larger than the median

(b) For left-skewed distributions, how will the mean compare to the median?
- ◯ The mean will be smaller than the median
- ◯ The mean will be larger than the median

Variability

As discussed above, there are two main modes of variability we use in this class: the SD and the IQR. For this course, we will use the following convention:

- When the data is skewed, we will use the IQR to describe the spread
- When the data is symmetric, we will use the SD to describe the spread.

⚠ Caution

Note that for symmetric distributions the SD and IQR may be different from one another.

Outliers

We use the term "outlier" to refer to any observation that is somewhat unusual. On a histogram, this manifests as points that seem to be far away from the bulk of the plot. For example, in Figure (5) above we see that most of the prices lie between 500 and around 6500 per month, but there are a handful of apartments with rents in the 7250-8000 range. These apartments would be considered outliers.

⚠ Caution

> Even if there aren't any outliers in a histogram, you should explicitly say that! If you don't, it looks like you didn't check for outliers.

We say that a statistic is **robust** if its value doesn't change significantly with the inclusion of outliers. Though this isn't a formal definition of robustness, we can still use it to gain some intuition:

Question 14

Consider the list of numbers $\{1, 2, 3, 4, 5\}$, and suppose we incorporate another element to create the list $\{1, 2, 3, 4, 5, 1000\}$.

(a) By how much does the mean change after incorporating the element 1000?

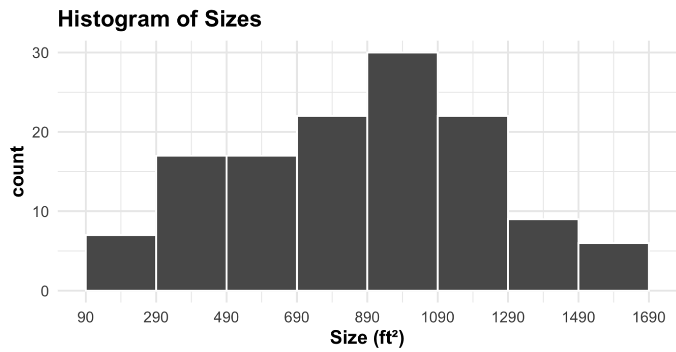
(b) By how much does the median change after incorporating the element 1000?

(c) Based on these answers, do you think the mean is a robust statistic? What about the median?

Another useful fact is that the IQR is a robust statistic, whereas the SD is not.

Putting It Together

It's time for you to put everything together! Shown below is a histogram of the **Size** variable along with some summary statistics.



Q_1	566.00
Mean	878.50
Median	900.00
Q_3	1141.50
SD	368.93

Figure 8: A histogram of apartment sizes.

⚠ Caution

When describing the center and variability of a distribution, you should explicitly state which measure (mean/median and SD/IQR) you are using and why. This is especially true on exams!

Question 15

Describe the distribution of apartment sizes. Be sure to describe all four elements: shape, center, variability, and outliers and use correct statistical terminology wherever appropriate.

Raving Restaurant Reviews

Suppose customers have been asked to rate four restaurants (named A, B, C, and D) on a scale of 1 ("loved it") to 9 ("hated it"). The results of this (fictional) survey are displayed below:

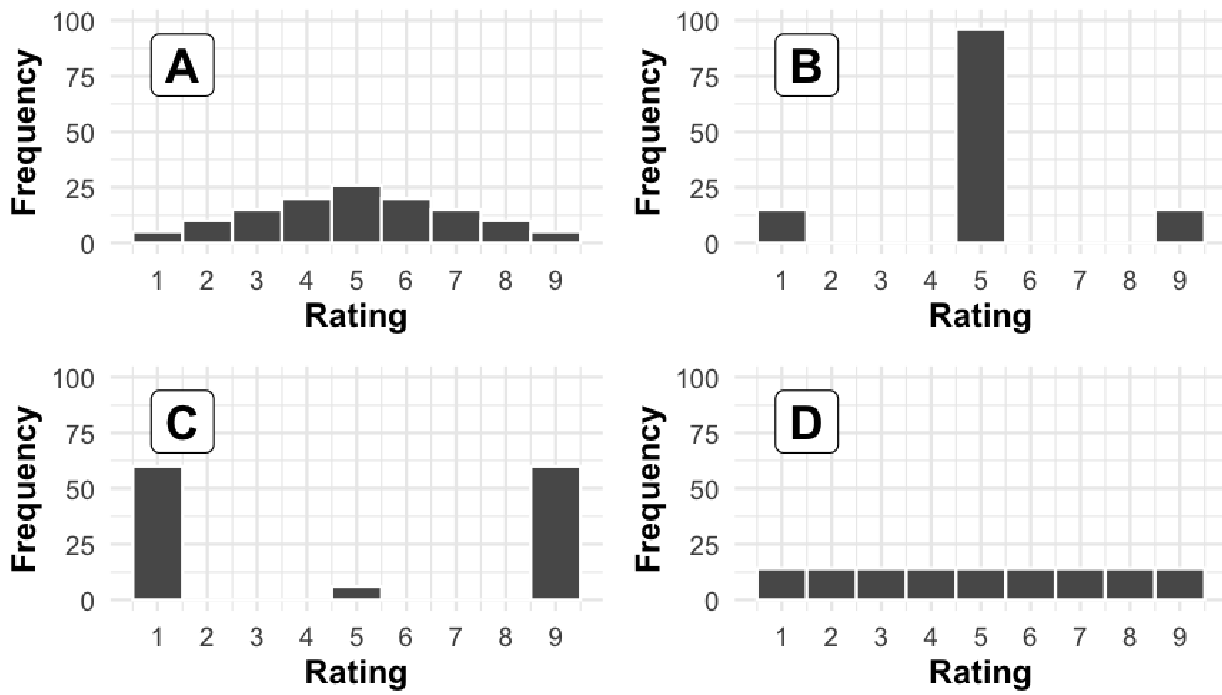


Figure 9: Ratings for four restaurants, on a scale of 1 ("hated it") to 9 ("loved it").

Question 16
How do the mean and median ratings compare across these four restaurants?

Question 17
Arrange these four restaurants in order from least variability (smallest SD) in rating scores to most variability (largest SD) in rating scores. **Hint:** least variability among the rating scores mean the most agreement/consistency among the raters; most variability means the least agreement/consistency.